

Digit Recognition using GMM and HMM

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Overview

The objective of this project is to build a system that can recognize spoken digits (0–9) from audio recordings. The dataset consists of 3000 recordings from multiple speakers.

To evaluate generalization:

- Training is performed on speakers 1–5
- Testing is performed on speaker 6 (unseen)

Two probabilistic models were implemented from scratch: **GMM** and **HMM**.

Feature Representation

We use **MFCC (Mel-Frequency Cepstral Coefficients)** to represent audio signals.

MFCC captures perceptually important frequency characteristics of speech. Each audio file is converted into a sequence of 13-dimensional feature vectors.

This representation allows the models to distinguish between different spoken digits based on spectral patterns.

Modeling Approaches

Gaussian Mixture Model (GMM) models each digit as a mixture of Gaussian distributions. It assumes that MFCC frames are independent and ignores temporal relationships.

Hidden Markov Model (HMM) models each digit as a sequence of hidden states. It captures how speech evolves over time, making it suitable for sequential data.

Experimental Setup

- Training data: Speakers 1–5
- Validation split: 80/20 from training data
- Test data: Speaker 6

Performance Comparison

Model	Validation Accuracy	Test Accuracy
GMM	0.84	0.50
HMM	0.804	0.756

Visual Results and Interpretation

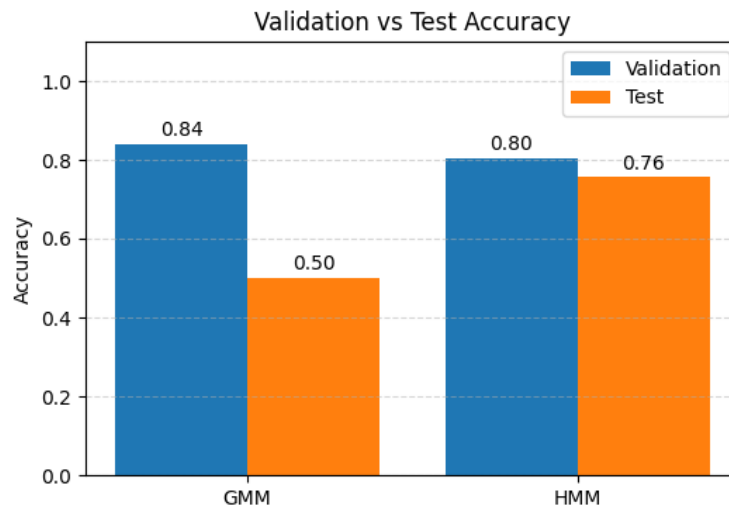


Figure 1: Validation vs Test Accuracy (X-axis: Model, Y-axis: Accuracy)

Insight:

GMM achieves higher validation accuracy than HMM but performs significantly worse on test data. This indicates that GMM is overfitting to the training speakers.

HMM, on the other hand, shows consistent performance across validation and test sets. This suggests that HMM generalizes better to unseen speakers due to its ability to model temporal dependencies.

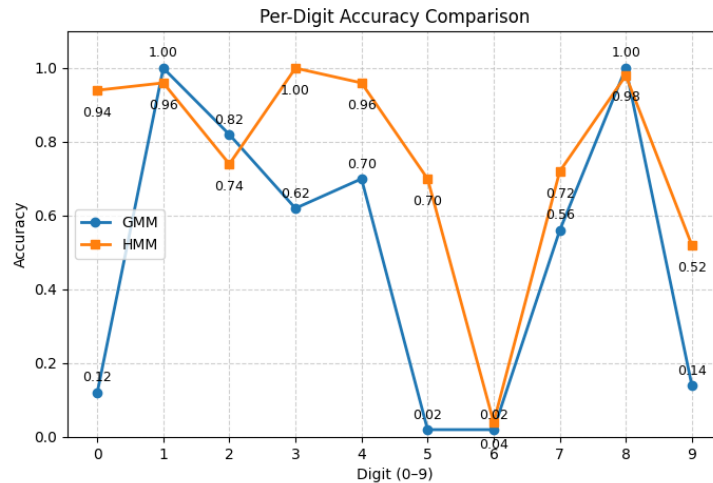


Figure 2: Per-Digit Accuracy (X-axis: Digit, Y-axis: Accuracy)

Per-Digit Performance

Digit	GMM	HMM
0	0.12	0.94
1	1.00	0.96
2	0.82	0.74
3	0.62	1.00
4	0.70	0.96
5	0.02	0.70
6	0.02	0.04
7	0.56	0.72
8	1.00	0.98
9	0.14	0.52

Insight:

HMM outperforms GMM across most digits, especially for digits like 0, 3, and 4, where temporal patterns are strong.

GMM performs well for simpler digits like 1 and 8 but fails for digits such as 5, 6, and 9. This is because these digits have overlapping MFCC distributions, which GMM cannot distinguish effectively.

Digit 6 performs poorly in both models, indicating that its acoustic features are highly similar to other digits.

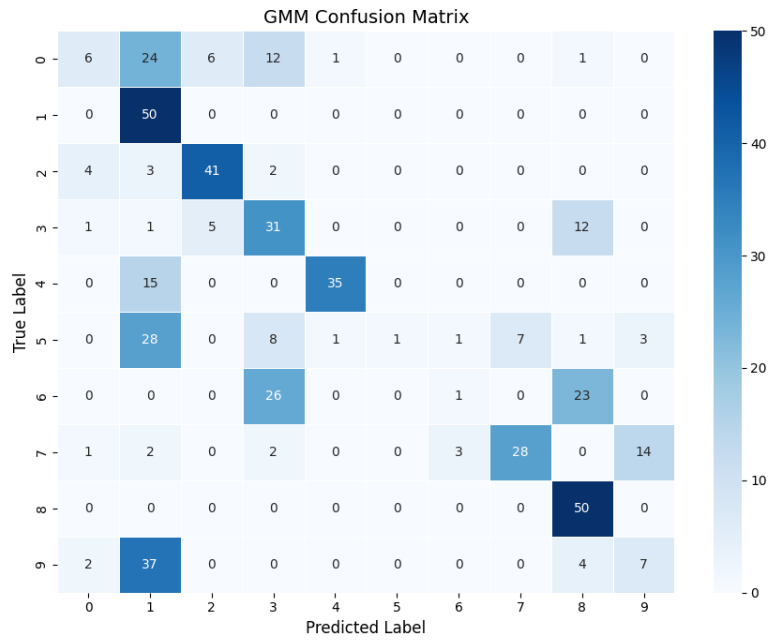


Figure 3: GMM Confusion Matrix (X-axis: Predicted, Y-axis: Actual)

Insight:

The GMM confusion matrix shows strong misclassification patterns. Digits such as 5 and 9 are frequently predicted as 1, indicating that GMM struggles with overlapping feature distributions.

This behavior highlights the limitation of GMM, as it does not consider temporal relationships between MFCC frames.

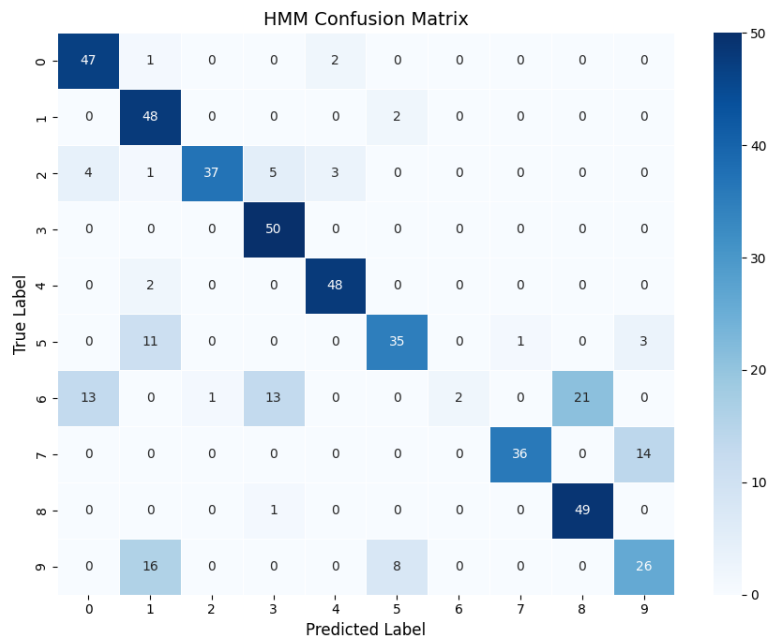


Figure 4: HMM Confusion Matrix (X-axis: Predicted, Y-axis: Actual)

Insight:

The HMM confusion matrix shows significantly fewer misclassifications compared to GMM. Most digits are correctly classified, with strong diagonal dominance.

This demonstrates that HMM effectively captures the temporal structure of speech, leading to improved discrimination between digits.

Model Tuning and Observations

GMM Tuning

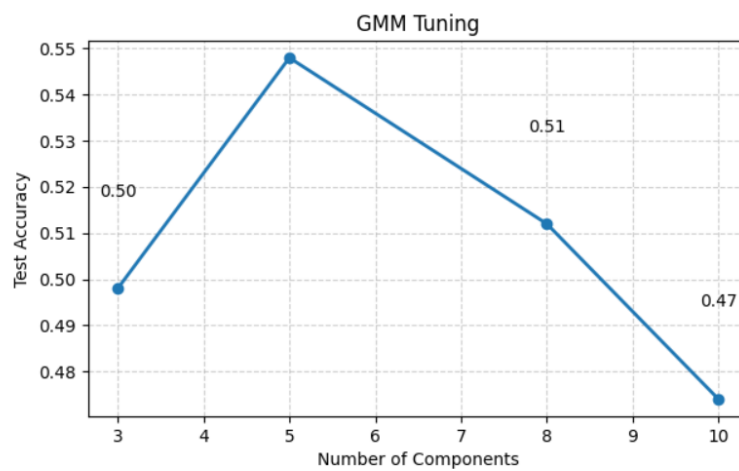


Figure 5: GMM Performance vs Number of Components (X-axis: Components, Y-axis: Accuracy)

Insight:

Increasing the number of Gaussian components improves validation accuracy initially, but test accuracy does not improve and may even decrease.

This indicates that higher model complexity allows GMM to fit training data more closely, but reduces its ability to generalize to unseen speakers. This is a clear sign of overfitting.

HMM Tuning

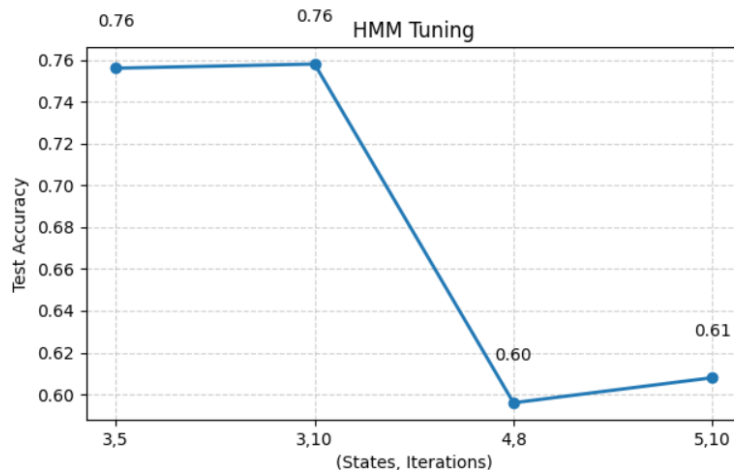


Figure 6: HMM Performance vs Number of States (X-axis: States, Y-axis: Accuracy)

Insight:

Increasing the number of states improves validation performance up to a point, but test accuracy decreases when the model becomes too complex.

This shows that while additional states allow HMM to model finer temporal patterns, too many states lead to overfitting. The best performance is achieved with a moderate number of states.

Model Behavior and Insights

GMM achieves high validation accuracy but fails on test data, indicating overfitting. HMM performs consistently across validation and test sets due to its ability to model temporal dependencies in speech.

Overfitting in GMM

GMM achieves strong performance on validation data but fails to generalize to unseen speakers. This happens because it models MFCC features independently and learns speaker-specific patterns instead of digit-specific characteristics.

Generalization of HMM

HMM maintains consistent performance across validation and test sets. It captures transitions between states, allowing it to model the temporal structure of speech effectively.

Confusion Analysis

GMM frequently confuses digits such as 5 and 9 with digit 1 due to overlapping feature distributions. HMM significantly reduces such confusion by incorporating temporal dependencies.

Difficult Digits

Digit 6 performs poorly in both models. This suggests that its acoustic features overlap with other digits, making it difficult to distinguish.

Conclusion

HMM is more suitable for this task because it captures temporal dependencies in speech signals, leading to better generalization and higher test accuracy.

GMM, while effective in modeling feature distributions, fails to capture sequential patterns and suffers from overfitting.